
Probabilistic Battery Cycle-Life Prediction and Degradation-Aware Fast-Charging Optimization

Problem Statement and Project Charter

A quantitative / embedded-ML research project on EV fast-charging battery health — fully software-based, no laboratory or hardware required.

Project type	Quantitative / Embedded Machine-Learning Research (prediction under uncertainty + risk-constrained optimization)
Application domain	EV fast charging, lithium-ion battery health, power-electronic charging systems
Host research area	Power-electronic converters for EV charging — research group of Dr. Siba Kumar Patro, Department of Electrical Engineering, IIT Roorkee
Candidate intern	Aditya Raj, BS-MS, IIT Roorkee (Markets / Quantitative Analysis and Data Science)
Defensible titles	“Quantitative Research Intern” and “Embedded ML Research Intern” (see Section 4.1)
Environment	Public datasets only; reproducible Python pipeline; no lab access or physical hardware needed
Document	Version 1.0

1. Project Overview

This project builds a probabilistic machine-learning system that predicts the cycle life and state-of-health (SoH) of a lithium-ion battery from its early-cycle measurements under a specified fast-charging protocol, and reports that prediction with a calibrated uncertainty band rather than a single number. The same model then drives a decision layer that recommends a charging-current protocol minimising charging time subject to an explicit, probabilistic battery-life constraint. The system is trained on the Toyota Research Institute (TRI) fast-charging dataset and validated on the CALCE and NASA battery datasets — all public and downloadable, with no laboratory work involved.

The work has a deliberate dual purpose. For the intern it is a rigorous, end-to-end quantitative and embedded-ML portfolio project covering probabilistic forecasting, uncertainty quantification, risk-constrained optimisation, survival-style modelling, explainability and a deployable model. For the host research group it supplies the battery-health prediction and charging-reference logic that feed the group's power-electronic fast-charging converters — a layer the hardware-focused team does not normally build itself.

2. Introduction and Background

Electric-vehicle adoption is driving demand for fast charging, which creates a fundamental engineering dilemma: charging faster (higher current) accelerates battery degradation. Battery ageing is nonlinear, cell-to-cell variable and sensitive to the exact charging profile, and first-principles or datasheet models of it are imprecise. This makes data-driven prediction — crucially, prediction that quantifies its own uncertainty — genuinely valuable.

The TRI / Severson study showed that early-cycle data alone can accurately predict a cell's eventual cycle life, and the follow-up closed-loop work (Attia et al.) used such a predictor to automatically select high-cycle-life fast-charging protocols. Together they establish the predict-then-decide arc this project follows: first predict life under uncertainty, then optimise the charging decision against it.

Relationship to the host group. Dr. Patro's group designs power-electronic converters and their control for EV fast charging — multiport charging stations, closed-loop constant-current/constant-voltage (CC-CV) control of dual-active-bridge converters, and hybrid silicon / silicon-carbide converters. That hardware physically delivers the charging current; this project supplies the battery-health prediction and the charging-reference logic that should govern it. The project therefore complements the group's hardware work by filling a

machine-learning and decision-making gap, without claiming to be converter design itself.

3. Problem Statement

Given early-cycle measurements of a lithium-ion cell charged under a specified fast-charging protocol, the project must: (a) predict the cell's cycle life / state-of-health together with a calibrated uncertainty interval; and (b) recommend a charging-current protocol that minimises charging time subject to a probabilistic battery-life constraint (for example, "at least $L_{\min}$ cycles with 90% confidence"). The solution must be fully reproducible from public data, require no laboratory hardware, and yield a compact predictor suitable for on-controller (battery-management / DSP) deployment.

4. Objectives

4.1 Intern's objectives (skills and career)

- Demonstrate end-to-end probabilistic regression / time-series modelling with **calibrated uncertainty** (conformal prediction or quantile regression), not just point estimates.
- Demonstrate **risk-constrained stochastic optimisation** and **survival / time-to-failure modelling** — methods that transfer directly to quantitative, risk and actuarial roles (time-to-failure is mathematically the time-to-default object).
- Produce a **compact, deployable SoH estimator** intended for on-controller use — the textbook embedded-ML artefact.
- Add **explainability** (SHAP) and a stakeholder-facing dashboard — the data-science / analyst signal.
- Ship a **reproducible, GitHub-ready repository** with a narrating notebook that can be walked through in an interview.
- Support two defensible internship titles: **Quantitative Research Intern** (probabilistic prediction under uncertainty plus risk-constrained optimisation) and **Embedded ML Research Intern** (the on-controller SoH estimator). Both describe the methodology, which is genuinely quantitative and deployable.

4.2 Research objectives aligned with Dr. Patro's group

- **Charging-protocol recommendations** that preserve battery life, usable as a health-aware basis for prioritising power across ports in a multiport fast-charging station (relevant to scholar **Jayant Kumar**, fast-charging infrastructure).
- An **SoH-adaptive current reference / setpoints** for closed-loop CC-CV control of the dual-active-bridge converter, plus a compact SoH estimator the controller can track as cells age (relevant to scholar **Dhayapule Mounika**, CC-CV DAB control).
- **Charging-stress and operating-point profiles** that inform the thermal and operating-point design of the hybrid Si-SiC EV-charging converter (relevant to the Infineon-funded converter project).
- A concrete **ML-on-controller SoH example** aligned with the teaching of Digital Control Implementation for Power Converters (EEN-635).
- Positioning for a **possible co-authored contribution** at the group's publication venues (SEFET / ECCE), where battery fast-charging life-prediction studies do appear.

4.3 Honest scope and ceiling

This is battery-side data science and charging-decision logic — not power-electronic converter topology design. Its value to the host group is as a complementary input/decision layer that the hardware-first team does not build. Treated that way it is genuinely useful and, realistically, a modest shared publication is plausible; it should not be presented as converter research.

5. Datasets and Resources (with access links)

All data is public and reachable online. The recommended access route is **BatteryML**, which bundles the major sets in a standardised, preprocessed form and removes per-dataset parsing. Note (verified at time of writing): the primary TRI host returns an intermittent 503, and NASA's own direct download is temporarily unavailable — both are covered by the mirrors listed below.

Dataset	Role	Access link	Notes
TRI / MATR (Severson + Attia)	Primary — training and test	https://data.matr.io/1/projects/5c48dd2bc625d700019f3204	124+ LFP cells, 72 fast-charge protocols, 30°C. Primary host intermittently 503 — prefer BatteryML / mirror.
BatteryML (Microsoft)	Recommended access + preprocessing	https://github.com/microsoft/BatteryML	Bundles MATR, CALCE, NASA, HUST, HNEI, SNL, UL_PUR with download scripts and a unified format.
CALCE (Univ. of Maryland)	Validation — other chemistries	https://calce.umd.edu/battery-data	CS2 / CX2 prismatic LCO cells, CC-CV; widely used for SoH and remaining-useful-life work.
NASA PCoE battery set	Validation	nasa.gov — PCoE data-set repository	Official page; direct download temporarily unavailable. Use mirrors below.
NASA mirror (Kaggle)	Validation (mirror)	https://www.kaggle.com/datasets/patrickfleith/nasa-battery-dataset	Requires a Kaggle API token to download programmatically.
NASA mirror (Zenodo)	Validation (mirror)	https://doi.org/10.5281/zenodo.15277374	Randomised Battery Usage set; direct DOI download, no token needed.
Severson et al. (2019)	Key reference	https://doi.org/10.1038/s41560-019-0356-8	Nature Energy — early-cycle cycle-life prediction (the dataset paper).
Attia et al. (2020)	Key reference	Nature 578, 397–402 (2020)	Closed-loop optimisation of fast-charging protocols — the predict-then-decide precedent.

Access strategy: the analysis sandbox can fetch all of the above programmatically (Python + internet), so no manual upload is required; CALCE, GitHub, Hugging Face and Zenodo are reachable. Because the sandbox is ephemeral, the durable artefact is a repository with download scripts that re-fetch the data on each run. Optional supplementary public sets reachable via BatteryML include HUST, Sandia (SNL) and HNEI.

6. Methodology and Technical Approach

A single predict-then-decide pipeline organised in four stages.

Stage 0 — Data foundation

Ingest the TRI/MATR data (via BatteryML) and the validation sets. Engineer early-cycle features: the cycle-to-cycle change in the discharge capacity-voltage curve (the variance and minimum of this signal are the strongest known predictors), capacity-fade slope, internal-resistance trend, temperature, charge time, and the protocol parameters. Use the canonical train / primary-test / secondary-test splits.

Stage 1 — Probabilistic life and SoH predictor (core)

Build a regularised linear / elastic-net baseline on engineered features (Severson-style), a gradient-boosted-tree model (LightGBM), and a sequence model (LSTM or 1D-CNN) on the cycle curves, then ensemble them. Wrap the ensemble with split-conformal prediction or quantile regression to produce calibrated prediction intervals, and report both interval coverage and interval width. Also export a compact, low-footprint version of the model for on-controller deployment.

Stage 2 — Risk-constrained charging-protocol optimisation (decision)

Define the space of admissible charging protocols. Objective: minimise predicted charging time. Constraint: the model's lower predictive bound on life at confidence (1 minus alpha) must meet a target $L_{\min}$ (a CVaR-style / quantile constraint). Search the protocol space with Bayesian optimisation and output both the recommended protocol(s) and the full speed-versus-life Pareto frontier.

Stage 3 — Communication and explainability

Compute global and local SHAP feature attributions, and build a dashboard (Streamlit) showing predictions with uncertainty bands, protocol comparisons, the Pareto frontier, and the feature importances.

Formulation, in words. (1) The predictor outputs a life estimate with an interval calibrated so the true life falls inside it with the stated probability on held-out cells. (2) The optimiser solves: minimise charge-time(protocol) subject to lower-life-bound(protocol, confidence) $\geq L_{\min}$, over the protocol space.

7. Key Features

- Probabilistic cycle-life / SoH prediction with **calibrated confidence intervals**.
- Accurate **early prediction** from roughly the first 100 cycles, before visible degradation.
- An **ensemble** of an interpretable feature model and a deep sequence model.
- A **risk-constrained charging-protocol recommender** plus a speed-versus-life Pareto frontier.
- A **compact, embeddable SoH estimator** suitable for a battery-management controller / DSP.
- **SHAP explainability** for every prediction.
- An **interactive dashboard** for non-technical stakeholders.
- A **config-driven, reproducible pipeline** with cross-dataset validation (train on one set, test on another).

8. Deliverables

Pipeline deliverables

- **Stage 0** — data loaders (TRI/MATR via BatteryML; CALCE and NASA validation), feature-engineering module, reproducible splits.
- **Stage 1** — baseline + ensemble predictor; conformal / quantile uncertainty layer; calibration and coverage report; compact on-controller SoH estimator; metrics report.
- **Stage 2** — risk-constrained optimisation module; recommended fast-charging protocol(s); speed-versus-life Pareto frontier.
- **Stage 3** — interactive dashboard; SHAP explainability panel.

Supporting artefacts

- A **GitHub-ready repository** (source, tests, README, requirements) that reproduces everything end-to-end.
- A **narrating Jupyter notebook** for interview walkthroughs.
- A **one-page research note** written in the host group's vocabulary, to secure supervisor buy-in.
- **Publication-style result figures** suitable for a SEFET / ECCE submission.

Priority note: Stages 0 and 1 form a complete, defensible project on their own and should be finished first. Stage 2 is the most valuable but also the hardest part; if time is short before placement it becomes the “and I extended it to...” rather than a blocker.

9. Evaluation and Metrics

Aspect	Metric	Target / definition
Point accuracy	RMSE, MAE, MAPE on cycle life	Lower is better; competitive with the published benchmark (single-digit percent error on the primary test set).
Uncertainty — calibration	Prediction-interval coverage (PICP)	Empirical coverage matches the nominal level (e.g. a 90% interval covers about 90% of held-out cells).
Uncertainty — sharpness	Mean interval width (MPIW)	As narrow as possible at the target coverage.
Early prediction	Error vs number of early cycles used	Accurate predictions from roughly the first 100 cycles.
Optimisation	Charge-time reduction %, life gain %, constraint satisfaction	Recommended protocols Pareto-dominate baseline protocols; life constraint met at the stated confidence.
Explainability	SHAP attribution sanity	Top features are physically sensible (e.g. capacity-curve variance).
Robustness	Cross-dataset transfer	Train on MATR, test on CALCE / NASA with graceful degradation.

10. Expected Results and Success Criteria

Success means: (1) a predictor whose accuracy is competitive with the published TRI benchmark and whose prediction intervals are demonstrably calibrated (coverage close to nominal); (2) a recommended fast-charging protocol that measurably reduces charging time relative to a baseline while honouring the battery-life constraint at the chosen confidence, supported by a clear Pareto frontier; (3) a compact estimator small enough to be credible as an on-controller model; and (4) a clean, reproducible repository plus the one-page note that lets the supervisor see the utility at a glance. Hitting (1) and (2) alone already constitutes a strong, defensible outcome.

11. Extensions and Future Work

- **Reinforcement-learning online charging control** — replace the static optimiser with a learned policy that adjusts current in real time as the cell ages.
- **Multiport power allocation under demand uncertainty** — allocate a station's limited converter capacity across ports using the health-aware predictions (direct tie to the multiport-station work).
- **Closed-loop integration with a converter simulation** (PLECS / Simulink) — feed the optimised current reference into a CC-CV dual-active-bridge model.
- **On-device deployment** — quantise and benchmark the compact estimator for a DSP / microcontroller (TinyML).
- **Chemistry transfer** — extend across LCO / NMC cells (CALCE) and explore transfer / foundation-model approaches.
- **Co-authored conference paper** at SEFET / ECCE summarising the prediction-plus-protocol results.

12. Tools and Technology Stack

Layer	Tools
Language and data	Python 3.x, NumPy, pandas; BatteryML for dataset access and preprocessing
Modelling	scikit-learn, LightGBM / XGBoost; PyTorch (LSTM / 1D-CNN) for the sequence model
Uncertainty	MAPIE or crepes (conformal prediction); quantile regression as an alternative
Optimisation	Optuna / scikit-optimize / BoTorch (Bayesian optimisation over the protocol space)
Explainability and UI	SHAP; Matplotlib / Plotly; Streamlit for the dashboard
Engineering	Git / GitHub, Jupyter, config files for reproducibility

Reference links: MAPIE · crepes · SHAP · LightGBM · Optuna · Streamlit

13. Suggested Timeline and Milestones

Week	Milestone
1	Environment setup; Stage 0 data pipeline via BatteryML; exploratory analysis of the TRI cells.
2–3	Baseline feature model and ensemble predictor (Stage 1 core); first accuracy metrics.
4	Conformal / quantile uncertainty layer; calibration and coverage report.
5	Cross-dataset validation (CALCE / NASA); SHAP; export the compact estimator.
6	Risk-constrained protocol optimisation and Pareto frontier (Stage 2).
7	Dashboard (Stage 3) and publication-style figures.
8	Repository polish, narrating notebook, one-page supervisor note; optional paper draft.

14. References and Links

- K. A. Severson et al., “Data-driven prediction of battery cycle life before capacity degradation,” *Nature Energy*, 2019. <https://doi.org/10.1038/s41560-019-0356-8>

- P. M. Attia et al., "Closed-loop optimization of fast-charging protocols for batteries with machine learning," Nature, 2020.
<https://doi.org/10.1038/s41586-020-1994-5>
- TRI fast-charging dataset: <https://data.mtr.io/1/projects/5c48dd2bc625d700019f3204>
- BatteryML (Microsoft): <https://github.com/microsoft/BatteryML>
- CALCE battery data (Univ. of Maryland): <https://calce.umd.edu/battery-data>
- NASA PCoE data-set repository:
<https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/>
- NASA battery mirrors: Kaggle ; Zenodo DOI

Prepared as a working problem statement. Dataset availability notes (TRI 503, NASA direct download) reflect the state verified at the time of writing and may change; the listed mirrors and BatteryML provide robust alternative access.